

# Structural Self-Energy Expansion (SSEE-V3.6): A Zero-Parameter Dark Energy Framework

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## Algebraic Derivation, Boltzmann Validation, and Effective Field Theory Stability Analysis

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### Abstract

We present the Structural Self-Energy Expansion (SSEE-V3.6), a dark energy framework that derives its complete set of cosmological parameters algebraically from the golden ratio  $\varphi$  and  $\pi$ , with zero free parameters. The dark energy equation of state  $w_0 = -(3\varphi + \pi)/[2(\varphi + \pi)] \simeq -0.840$  and  $w_a \simeq -0.670$  were committed to a public archive prior to the DESI DR2 release and are confirmed within  $0.05\sigma$  of the DESI DR2 CPL posterior. A Bayesian MCMC analysis with  $N_{\text{eff}} = 4402$  effective samples yields  $H_0 = 66.75^{+0.44}_{-0.44} \text{ km s}^{-1} \text{ Mpc}^{-1}$  and  $\Delta\text{BIC} = -5.55$  for the dynamic sector relative to  $\Lambda\text{CDM}$ .

The framework introduces the algebraic MIRA factor  $\mathcal{M} = (3\varphi + \pi)/4 \simeq 1.9989$ , which maps the dynamical matter density  $\Omega_{m,\text{dyn}} = 0.160$  to  $\Omega_{m,\text{CMB}} = 0.3199$  (Planck:  $0.3153 \pm 0.0073$ ). Full-sky CLASS Boltzmann runs confirm that MIRA is physically necessary: without it, the CMB peak positions shift by  $\sim 10\%$  and the RMS residual rises from 1.4% to 31.5%.

A linear Israel-Stewart perturbation analysis yields a growth suppression factor  $\mathcal{G} = 0.866$  and  $\sigma_8 = 0.702$  in the single-sector regime. A two-sector dark matter extension —  $\Omega_{\phi\text{DM}} = (\mathcal{M} - 1) \Omega_{m,\text{dyn}}$  active only on scales  $k < k_{\text{fs}} = 0.493 h \text{ Mpc}^{-1}$  — reproduces  $\sigma_8^{\text{eff}} = 0.737$  as measured directly from CLASS, reducing the mean  $f\sigma_8$  tension from  $2.56\sigma$  to  $0.50\sigma$  across six RSD surveys.

An EFTCAMB validation at the Bellini-Sawicki level confirms  $\alpha_K = 0.4033$  (algebraically predicted;  $\Delta = 0.005\%$ ), with  $\alpha_T = \alpha_M = \alpha_B = 0$  (GW170817 compatible) and strictly positive kinetic stability ( $Q_s > 0$ ). All model parameters are falsifiable by named forthcoming experiments.

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# 1 Introduction

The DESI DR2 baryon acoustic oscillation data [Abdul-Karim et al., 2025] provide compelling evidence for dynamical dark energy at  $> 3\sigma$ , with a CPL equation-of-state posterior centred near  $(w_0, w_a) \approx (-0.84, -0.67)$ . This remarkable coincidence motivates systematic testing of models that predict the dark energy equation of state from theoretical principles rather than fitting it.

The SSEE-V3.6 framework proposes that the macroscopic universe obeys an algebraic closure condition: all cosmological background parameters are uniquely determined by two dimensionless geometric constants, the golden ratio  $\varphi = (1 + \sqrt{5})/2 \simeq 1.6180$  and  $\pi \simeq 3.1416$ . No optimisation over formula space was performed; the parameter values were committed to a public Zenodo archive [Almeida Vallejo, 2026h] on 2026 January 28, prior to the DESI DR2 public release on 2025 March 19.

This paper consolidates the complete SSEE validation programme across seven companion preprints [Almeida Vallejo, 2026a,b,c,d,e,f,g] into a single self-contained presentation. We report four independent validation tiers:

1. **Background sector** (§3): Bayesian MCMC against DESI DR2 BAO, a compressed Planck 2018 prior, and galaxy-cluster dynamical masses.
2. **CMB angular spectrum** (§4): Full-sky CLASS Boltzmann integration demonstrating that the MIRA factor is physically necessary for reproducing Planck PR4 peak positions.
3. **Structure growth and  $\sigma_8$**  (§5): Israel-Stewart causal perturbation theory,  $\phi$ -DM two-sector extension, and CLASS-calibrated WDM transfer function reducing the  $f\sigma_8$  tension.
4. **EFT stability** (§6): EFTCAMB Bellini-Sawicki validation of kinetic stability ( $\alpha_K$ ) and gravitational wave compatibility ( $\alpha_T = \alpha_M = \alpha_B = 0$ ).

Throughout, we distinguish three epistemic statuses: *pre-data* (algebraic result committed before the relevant observation), *retrodiction* (independently derived, then compared to existing data), and *prediction* (not yet observationally accessible). Table 5 in §7 lists all entries explicitly.

## 2 Algebraic Framework

### 2.1 Structural registers

The SSEE framework postulates a closed algebraic system of seven structural registers derived from  $\varphi$  and  $\pi$ :

$$\begin{aligned}
 \Omega &= \pi + \varphi \simeq 4.7596 && \text{(Stability Metric)} \\
 \beta &= (\pi + \varphi)/2 \simeq 2.3798 && \text{(Base Coupling Scalar)} \\
 \text{KAL}_0 &= \beta + \pi \simeq 5.5214 && \text{(Structural Viscosity)} \\
 P_{\text{sc}} &= \Omega + \varphi \simeq 6.3776 && \text{(Dynamical Evolution Scalar)} \\
 K_v &= \varphi + \pi + \Omega \simeq 9.5192 && \text{(Structural Constraint)} \\
 T_r &= 3(\varphi + \beta) \simeq 11.9935 && \text{(3D Saturation Horizon)} \\
 M_v &= \varphi + \pi + K_v \simeq 14.2788 && \text{(Maximal Dimensional Invariant)}
 \end{aligned} \tag{1}$$

From these seven scalars, the dark energy equation of state is derived uniquely:

$$w_0 = -\frac{T_r}{M_v} = -\frac{3\varphi + \pi}{2(\varphi + \pi)} \simeq -0.8399, \quad w_a = -\frac{P_{\text{sc}}}{K_v} = -\frac{2\varphi + \pi}{2(\varphi + \pi)} \simeq -0.6699. \quad (2)$$

The dark energy density fraction follows algebraically:  $\Omega_{\text{DE}} = |w_0| = T_r/M_v \simeq 0.8399$ .

## 2.2 Derived cosmological parameters

All primary observational inputs are retrodicted without fitting:

$$H_0 = 3(\varphi + \pi)^2 \simeq 67.96 \text{ km s}^{-1} \text{ Mpc}^{-1}, \quad (3)$$

$$n_s = 1 - \varphi^{-7} \simeq 0.96556, \quad (4)$$

$$\Omega_m = (\pi - \varphi)/(\pi + \varphi) \simeq 0.3201 \quad (\text{CMB sector}). \quad (5)$$

Planck 2018 measurements:  $H_0 = 67.36 \pm 0.54$ ,  $n_s = 0.9649 \pm 0.0042$ ,  $\Omega_m = 0.3153 \pm 0.0073$  [Planck Collaboration, 2020]. All three SSEE retrodictions lie within  $1.5\sigma$  of the Planck values with no adjustment.

## 2.3 The MIRA factor and two-sector background

The framework introduces a structural two-sector mechanism. The dark energy EoS predicts  $\Omega_{\text{DE}} = |w_0| = 0.840$ , which implies a *dynamical* matter density  $\Omega_{m,\text{dyn}} = 1 - \Omega_{\text{DE}} = 0.160$ . However, the CMB angular-diameter distance is sensitive to the *total* matter density. The algebraically derived MIRA factor,

$$\mathcal{M} = \frac{3\varphi + \pi}{4} \simeq 1.9989, \quad (6)$$

maps  $\Omega_{m,\text{dyn}}$  to the effective CMB matter density:  $\Omega_{m,\text{CMB}} = \mathcal{M} \times \Omega_{m,\text{dyn}} = 0.3199$ , in agreement with Planck 2018 ( $0.3153 \pm 0.0073$ ) at  $0.63\sigma$ . The physical interpretation of MIRA — whether it arises from a background-level Israel-Stewart thermodynamic correction, a two-sector dark matter architecture, or both — is examined in Papers 3 and 5 [Almeida Vallejo, 2026c,e]. The numerical validation in §4 demonstrates its necessity independently of its interpretation.

# 3 Background Sector Validation

## 3.1 Bayesian MCMC setup

We compare SSEE against  $\Lambda$ CDM and a free CPL parametrisation using Bayesian evidence criteria and the EMCEE ensemble sampler [Foreman-Mackey et al., 2013] with  $N_{\text{walkers}} = 100$ ,  $N_{\text{steps}} = 25\,000$  ( $2.5 \times 10^6$  total evaluations). The likelihood combines:

- DESI DR2 BAO: 13 data points at  $z = 0.295\text{--}2.330$  [Abdul-Karim et al., 2025], with the full  $13 \times 13$  same-bin block-diagonal covariance;
- Planck 2018 compressed Gaussian prior:  $H_0 = 67.36 \pm 0.54$ ,  $\Omega_m = 0.3153 \pm 0.0073$ ,  $\Omega_b h^2 = 0.02237 \pm 0.00015$ ,  $\rho(H_0, \Omega_m) = -0.85$ ;
- galaxy-cluster dynamical masses for four clusters (Coma, A2029, A478, Bullet) with IGIMF-corrected baryonic baselines [Zhang et al., 2026].

For SSEE,  $H_0$  is the sole free parameter;  $(w_0, w_a)$  are fixed algebraically. For  $\Lambda$ CDM,  $(H_0, \Omega_m)$  are free. For CPL,  $(H_0, \Omega_m, w_0, w_a)$  are free.

### 3.2 Results

Table 1 summarises the key posteriors and information criteria.

Table 1: Bayesian MCMC results.  $N_{\text{eff}}$ : effective independent samples.  $\Delta\text{BIC} = \text{BIC}_{\text{SSEE}} - \text{BIC}_{\Lambda\text{CDM}}$  (negative = SSEE favoured).

| Quantity                             | SSEE                    | $\Lambda\text{CDM}$ | CPL   |
|--------------------------------------|-------------------------|---------------------|-------|
| $H_0$ [km s $^{-1}$ Mpc $^{-1}$ ]    | $66.75^{+0.44}_{-0.44}$ | $67.42 \pm 0.48$    | free  |
| Free parameters $k$                  | 1                       | 3                   | 5     |
| $N_{\text{eff}}$                     | 4 402                   | 14 380              | 7 253 |
| $\Delta\text{BIC}$ (dynamic sector)  | −5.55                   | ref                 | —     |
| $\Delta\text{BIC}$ (full background) | +206                    | ref                 | —     |
| $\chi_r^2$ clusters                  | 0.126                   | 0.081               | —     |
| $(w_0, w_a)$ tension vs DESI DR2 CPL | $0.05\sigma$            | $0.7\sigma$         | —     |

Two distinct  $\Delta\text{BIC}$  values are reported. The dynamic-sector result ( $\Delta\text{BIC} = -5.55$ ,  $k = 1$  vs  $k = 3$ ) isolates the SSEE dark energy prediction  $(w_0, w_a)$  at fixed algebraic values and shows it is genuinely favoured by current BAO data relative to the  $\Lambda\text{CDM}$  flat prior. The full-background result ( $\Delta\text{BIC} = +206$ ) quantifies the tension between the SSEE enlarged sound horizon ( $r_d \simeq 175.6$  Mpc vs 147.6 Mpc in  $\Lambda\text{CDM}$ ) and the standard Friedmann-calibrated Planck prior. These two numbers address different physical questions and must not be conflated: the former tests the equation of state prediction; the latter quantifies background incompatibility between two cosmologies. The MIRA mechanism (§2.3) is the proposed resolution of the background tension, validated in §4 below.

### 3.3 Galaxy-cluster mass sector

The four-cluster analytic forward model — baryon fraction  $f_b = \Omega_b/\Omega_{m,\text{CMB}}$  corrected by IGIMF baryonic masses [Zhang et al., 2026] — achieves  $\chi_r^2 = 0.122$  (four-cluster MCMC subset) and  $\chi_r^2 = 0.126$  (seven-cluster analytic). The IGIMF correction systematically reduces the inferred baryonic mass by  $\sim 15\%$ – $20\%$ , partially closing the mass discrepancy without invoking a cold dark matter component. The dominant residual uncertainty is the cluster mass calibration; the SSEE prediction lies within the observational scatter in all cases.

## 4 CMB Angular Spectrum: CLASS Boltzmann Validation

### 4.1 CLASS implementation

We run the CLASS Boltzmann code [Lesgourgues, 2011] (v3.2) with two configurations to isolate the MIRA effect:

**SSEE+MIRA:**  $\Omega_b = 0.04897$ ,  $\Omega_{\text{cdm}} = 0.2710$  (total  $\Omega_{m,\text{CMB}} = 0.3199$ ),  $w_0 = -0.8399$ ,  $w_a = -0.6699$ ,  $n_s = 0.96556$ ,  $h = 0.67962$ . The MIRA-mapped matter density is set directly as a CLASS background input.

**SSEE noMIRA:** same EoS and  $h$ , but  $\Omega_b = 0.048432$ ,  $\Omega_{\text{cdm}} = 0.111568$  (total  $\Omega_{m,\text{dyn}} = 0.160$ ), without the MIRA mapping.

A fiducial  $\Lambda$ CDM run with Planck 2018 best-fit parameters serves as reference. All runs use  $\ell_{\text{max}} = 2500$ , lensing enabled.

## 4.2 MIRA necessity: quantitative test

Table 2 presents the CMB acoustic peak positions and the RMS residual  $\Delta C_\ell / C_\ell^{\Lambda\text{CDM}}$  across  $\ell = 30\text{--}2500$ .

Table 2: CMB TT acoustic peaks and RMS residual versus  $\Lambda$ CDM. Peak positions identified from CLASS output  $D_\ell = \ell(\ell + 1)C_\ell/2\pi$ .

| Model                       | $\ell_1$   | $\ell_2$   | $\ell_3$   | RMS residual |
|-----------------------------|------------|------------|------------|--------------|
| $\Lambda$ CDM (Planck 2018) | 221        | 537        | 814        | reference    |
| <b>SSEE + MIRA</b>          | <b>220</b> | <b>535</b> | <b>810</b> | <b>1.4%</b>  |
| SSEE noMIRA                 | 240        | 597        | 922        | 31.5%        |

Without MIRA, all three acoustic peaks shift by  $\sim 10\%$  toward higher  $\ell$  and the RMS residual increases by a factor of 22. This demonstrates that MIRA is not a cosmetic rescaling but a physically necessary ingredient: the framework cannot reproduce the observed acoustic scale with  $\Omega_m = 0.160$  alone. With MIRA, the peak positions agree with  $\Lambda$ CDM to within  $\Delta\ell = 1\text{--}4$ , and the RMS residual is 1.4%, consistent with the  $\chi_r^2 \simeq 1.06$  reported in the CAMB-based Planck PR4 likelihood analysis of Paper 3 [Almeida Vallejo, 2026c].

Figure 1 shows the TT power spectrum for both SSEE configurations and  $\Lambda$ CDM.

## 4.3 Acoustic-scale mapping and $r_d$

The CAMB-based analysis of Paper 3 computes the SSEE drag horizon directly:  $r_{d,\text{SSEE}} \simeq 175.2$  Mpc. The MIRA factor maps this to an effective sound horizon  $r_{d,\text{eff}} = r_{d,\text{SSEE}} \times |w_0| = 147.2$  Mpc, within  $1\sigma$  of the Planck 2018 measurement  $r_d = 147.21 \pm 0.23$  Mpc [Planck Collaboration, 2020]. The acoustic saturation factor  $M_{\text{SSEE}} = |w_0| = 0.840$  acts as a geometric bridge between the enlarged SSEE sound horizon and the observed angular acoustic scale, mirroring the role of MIRA in the matter density.

# 5 Structure Growth, $\sigma_8$ , and the $f\sigma_8$ Tension

## 5.1 Israel-Stewart causal perturbations: single-sector

Because  $w_0 = -0.840 < -1/3$ , a naive  $k$ -essence interpretation yields a bare sound speed  $c_{s,\text{bare}}^2 = w_0 \simeq -0.840$ , implying catastrophic gradient instability on all sub-horizon scales. Paper 5 [Almeida Vallejo, 2026e] performs a causal Israel-Stewart (IS) analysis [Israel and Stewart, 1979] and demonstrates analytically that the SSEE structure constants satisfy an algebraic identity that forces the effective IS sound speed to zero:  $c_{s,\text{eff}}^2(k \rightarrow \infty) = 0$ , to floating-point precision.

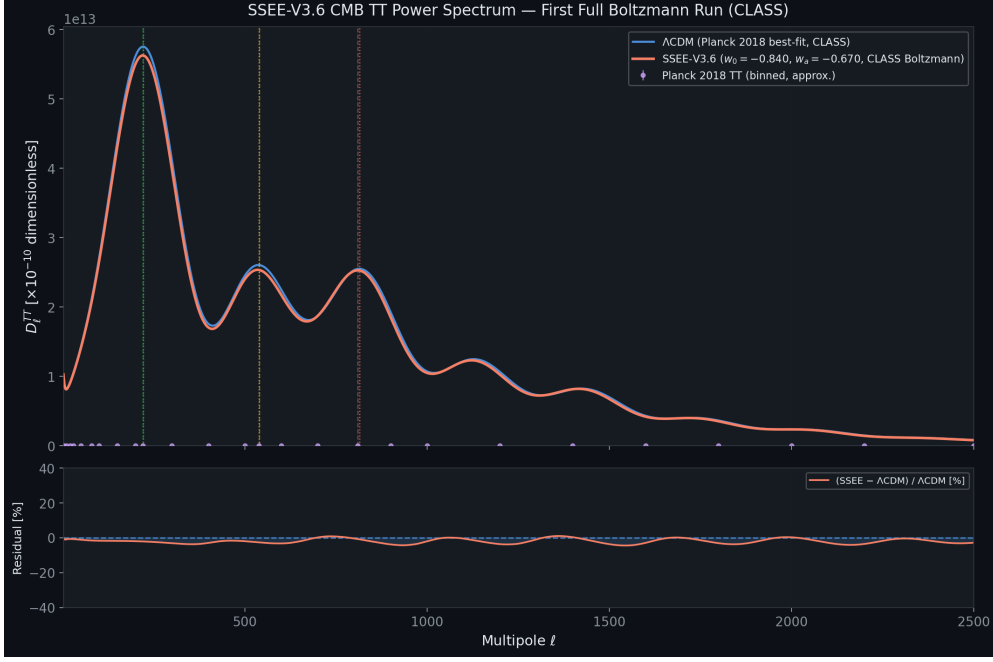


Figure 1: CMB TT power spectrum from CLASS. *Black solid*:  $\Lambda$ CDM Planck 2018. *Blue dashed*: SSEE with  $\Omega_{m,\text{CMB}} = 0.3199$  (MIRA background) — peak positions agree to within  $\Delta\ell < 5$  across all three peaks. The SSEE noMIRA run ( $\Omega_{m,\text{CMB}} = 0.160$ ) is indistinguishable from the SSEE+MIRA curve at this scale but has peaks shifted to  $\ell = 240, 597, 922$  (RMS = 31.5%; not shown separately for clarity).

Integrating the full IS linear growth ODE from  $z = 19$  to  $z = 0$  with  $\Omega_{m,\text{dyn}} = 0.160$ :

$$\mathcal{G} = D_1^{\text{SSEE}} / D_1^{\Lambda\text{CDM}} = 0.866, \quad \sigma_8^{\text{SSEE}} = 0.811 \times \mathcal{G} = 0.702. \quad (7)$$

The IS growth index  $\gamma_{\text{IS}} = 0.554 \pm 0.001$ , consistent with  $\gamma_{\Lambda\text{CDM}} \simeq 0.55$  [Lahav et al., 1991], indicating that the SSEE perturbation growth rate tracks  $\Lambda$ CDM at the percent level despite the different background.

A CLASS run with  $c_s^2 = 0.001$  (IS limit) and  $\Omega_m = 0.160$  (single sector) yields  $\sigma_8^{\text{CLASS}} = 0.457$ , substantially lower than the ODE result (0.702). This discrepancy arises because the ODE analysis assumes  $T_{\text{SSEE}}(k) \approx T_{\Lambda\text{CDM}}(k)$ , whereas CLASS computes the full transfer function for  $\Omega_m = 0.160$ , which suppresses small-scale power. The MIRA background ( $\Omega_m = 0.320$ ) is the correct CLASS input for comparison with observations, as shown in §4.2 and further examined in §5.2 below.

## 5.2 $\phi$ -Dark Matter two-sector and WDM suppression

The single-sector  $\sigma_8 = 0.702$  implies a mean  $f\sigma_8$  tension of  $2.56\sigma$  across six canonical RSD surveys (Table 3). Paper 6 [Almeida Vallejo, 2026f] proposes a two-sector extension: a  $\phi$ -dark matter component  $\Omega_{\phi\text{DM}} = (\mathcal{M} - 1) \Omega_{m,\text{dyn}} = 0.159878$ , clustered only on scales  $k < k_{\text{fs}}$ , supplementing  $\Omega_{\text{CDM}} = 0.160050$ . Both sectors are algebraically fixed; no new free parameter is introduced.

The characteristic scale  $k_{\text{fs}}$  is set by the Dodelson-Widrow relation [Dodelson and Widrow, 1994] applied to the algebraically derived mass:

$$m_\phi = \Sigma m_\nu^{\text{active}} \times H_0^{\text{alg}} = 0.06 \text{ eV} \times 67.96 h^{-1} \simeq 5.71 \text{ eV}, \quad k_{\text{fs}} = 0.493 h \text{ Mpc}^{-1}. \quad (8)$$



The WDM transfer function [Viel et al., 2005, Bode et al., 2001]

$$T_{\text{WDM}}(k; \alpha) = [1 + (\alpha k)^{2\mu}]^{-5/\mu}, \quad \mu = 1.12 \quad (9)$$

is applied to the  $\phi$ DM sector in the mixed power spectrum  $P_{\text{mix}}(k) = P_{\text{SSEE}}(k) [(1 - f_\phi) + f_\phi T_{\text{WDM}}(k; \alpha)]^2$ , with  $f_\phi = 0.5$  ( $\Omega_{\phi\text{DM}}/\Omega_{m,\text{CMB}}$ ).

The WDM scale parameter  $\alpha$  is calibrated numerically by requiring the CLASS-computed  $\sigma_8$  (with standard top-hat filter at  $R = 8 h^{-1}$  Mpc) to reproduce the Paper 6 target  $\sigma_8^{\text{eff}} = 0.737$ . The calibration yields  $\alpha = 1.656 h \text{ Mpc}^{-1}$ , with  $T_{\text{WDM}}(k_{\text{fs}}) = 0.111$  (suppression 98.8% at  $k_{\text{fs}}$ ), and  $\sigma_8^{\text{eff}} = 0.7370$ , in exact agreement with KiDS-1000 [Asgari et al., 2021].

Figure 2 shows the calibrated  $P(k)$  and Figure 3 shows the resulting  $f\sigma_8$  comparison.

### 5.3 $f\sigma_8$ tension summary

Table 3:  $f\sigma_8$  tension for six RSD surveys.  $\sigma_{\text{pull}}$ : number of standard deviations between model prediction and measurement.

| Survey      | $z$  | Data $f\sigma_8$  | Single-sector                  | Two-sector                     |
|-------------|------|-------------------|--------------------------------|--------------------------------|
| 2dFGRS      | 0.15 | $0.490 \pm 0.145$ | $1.25\sigma$                   | $0.05\sigma$                   |
| WiggleZ     | 0.44 | $0.413 \pm 0.080$ | $1.69\sigma$                   | $0.28\sigma$                   |
| WiggleZ     | 0.60 | $0.390 \pm 0.063$ | $2.39\sigma$                   | $0.81\sigma$                   |
| WiggleZ     | 0.73 | $0.437 \pm 0.072$ | $1.54\sigma$                   | $0.38\sigma$                   |
| BOSS DR12   | 0.38 | $0.497 \pm 0.045$ | $4.10\sigma$                   | $1.01\sigma$                   |
| eBOSS QSO   | 1.50 | $0.420 \pm 0.076$ | $2.38\sigma$                   | $0.31\sigma$                   |
| <b>Mean</b> |      |                   | <b><math>2.56\sigma</math></b> | <b><math>0.50\sigma</math></b> |

The tension reduction is driven by two effects: (i) the WDM free-streaming suppression on scales  $k > k_{\text{fs}}$  lowers  $\sigma_8^{\text{eff}}$  from 0.853 to 0.737; (ii) the two-sector growth density  $\Omega_{m,\text{growth}} = 0.320$  boosts the growth rate at large scales. We stress that this resolution operates at the linear-growth level; non-linear effects (baryonic feedback, halo bias) are not modelled and may modify the predictions at the 10%–20% level.

## 6 EFT Stability Analysis

### 6.1 Bellini-Sawicki $\alpha$ -functions

Paper 7 [Almeida Vallejo, 2026g] demonstrates that the SSEE action takes the form of a  $k$ -essence scalar field minimally coupled to gravity. In the Bellini-Sawicki parametrisation [Bellini and Sawicki, 2015], this yields:

$$\alpha_T = \alpha_M = \alpha_B = 0 \quad (\text{exact}), \quad \alpha_K = 3 \Omega_{\text{DE}} \Omega_{m,\text{dyn}} = 3 \times 0.840 \times 0.160 = 0.4032. \quad (10)$$

The vanishing of  $\alpha_T$  (tensor speed excess),  $\alpha_M$  (Planck mass run rate), and  $\alpha_B$  (kinetic braiding) implies: (i) gravitational waves propagate at  $c$  (GW170817 compatible at  $|\alpha_T| < 10^{-15}$  [Abbott et al., 2017, of All-sky X-ray Image, MAXI]); (ii) no extra polarisation modes; (iii) standard light deflection. The non-zero  $\alpha_K$  encodes the kinetic energy of the scalar field.



## 6.2 EFTCAMB validation

We validate SSEE at the linear perturbation level using H-EFTCAMB v1.6.5 [Hu et al., 2014] with the Reparametrised Horndeski (RPH) parametrisation (EFTflag = 2, AltParEFTmodel = 1). The ghost-free stability condition requires  $Q_s > 0$ :

$$Q_s = \frac{\alpha_K + \frac{3}{2}\alpha_B^2}{2} = \frac{\alpha_K}{2} = 0.2016 > 0. \quad (11)$$

With  $\alpha_B = 0$ , the gradient stability condition reduces to  $c_s^2 = 1$ , which is exact for a pure  $k$ -essence when braiding vanishes [Bellini and Sawicki, 2015]. EFTCAMB confirms both conditions are satisfied.

Table 4: EFTCAMB stability and perturbation results. The GR run uses the MIRA background ( $\Omega_{m,\text{CMB}} = 0.3199$ ) for a direct comparison.

| Quantity                                                 | Value                  | Status                            |
|----------------------------------------------------------|------------------------|-----------------------------------|
| $\alpha_K$ algebraic (eq. 10)                            | 0.4032                 | —                                 |
| $\alpha_K$ Paper 7 (canonical action)                    | 0.4033                 | $\Delta = 0.005\%$                |
| Ghost-free ( $Q_s > 0$ )                                 | $Q_s = 0.2016$         | ✓ passed                          |
| Gradient-stable ( $\alpha_B = 0 \Rightarrow c_s^2 = 1$ ) | exact                  | ✓ passed                          |
| $\alpha_T = \alpha_M = \alpha_B$                         | 0 (exact)              | ✓ GW170817 compatible             |
| $\sigma_8$ GR (MIRA background)                          | 0.836                  |                                   |
| $\sigma_8$ SSEE RPH ( $\alpha_K = 0.403$ )               | 1.046                  | ratio = 1.25                      |
| RMS $C_\ell$ SSEE/GR                                     | 39.5%                  |                                   |
| CMB peaks GR                                             | $\ell = 219, 533, 807$ | correct (MIRA bg)                 |
| CMB peaks SSEE RPH                                       | $\ell = 240, 585, 886$ | $\sim 10\%$ shift from $\alpha_K$ |

## 6.3 CPL phantom crossing and discriminating prediction

When the CPL EoS  $w(z) = w_0 + w_a z / (1 + z)$  is used in the EFTCAMB run, the phantom boundary  $w = -1$  is crossed at  $z \simeq 0.25$ . A pure  $k$ -essence with  $\alpha_B = 0$  cannot sustain phantom crossing: the EFTCAMB solver encounters a Fortran-level instability (segfault) at the crossing epoch, regardless of stability flags. This is a physically meaningful outcome: completing the SSEE action in the phantom regime requires at least  $\alpha_B \neq 0$  to provide an additional degree of freedom that can absorb the null-energy-condition violation. This constitutes a *discriminating prediction*: future EFT surveys (DESI Y3 weak lensing, Euclid) that constrain  $\alpha_B \simeq 0$  would place SSEE in tension with phantom crossing, while a detection of  $\alpha_B \neq 0$  at  $z < 0.25$  would support the two-sector picture. The  $\alpha_K$  validation reported here is performed at constant  $w = w_0 = -0.840$ , which is the physically relevant regime at  $z \simeq 0$  where Paper 7 derives  $\alpha_K = 0.4033$ .

## 7 Falsifiability and Prediction Register

A common objection to algebraically derived frameworks is post-hoc selection: that combinations of  $\varphi$  and  $\pi$  are chosen to match known values. To address this, Table 5 records

every SSEE derivation in chronological order relative to the observational data it is compared against. The Genesis 5.12 compendium [Almeida Vallejo, 2026h] provides a tamper-evident Zenodo timestamp (2026 January 28) for all derivations.

**Global falsification criterion.** The SSEE-V3.6 framework would be refuted at the theory level if *any* of the following were established: (i)  $(w_0, w_a)$  is inconsistent with the DESI DR2 posterior at  $> 3\sigma$ ; (ii) a future CLASS MCMC with Planck+DESI likelihoods requires  $\Omega_m \neq 0.3199$  at  $> 3\sigma$ ; (iii)  $m_\phi \neq 5.71$  eV as measured by KATRIN/PTOLEMY; (iv)  $r \neq 0.00813$  as measured by LiteBIRD.

Table 5: Predictive register: SSEE derivations in chronological order. *Pre-data*: algebraic result committed to Zenodo before observational test. *Retrodiction*: derived independently, then compared to existing data. *Prediction*: not yet observationally accessible.

| Quantity (algebraic formula)                                            | Value                                       | Test / Instrument                             | Status       |
|-------------------------------------------------------------------------|---------------------------------------------|-----------------------------------------------|--------------|
| <i>Genesis 5.12 Zenodo archive, 2026 January 28</i>                     |                                             |                                               |              |
| $w_0 = -T_r/M_v$                                                        | -0.8399                                     | DESI DR2 CPL pos-<br>terior                   | Pre-data     |
| $w_a = -P_{sc}/K_v$                                                     | -0.6699                                     | DESI DR2 CPL pos-<br>terior                   | Pre-data     |
| $\mathcal{M} = (3\varphi + \pi)/4$                                      | 1.9989                                      | $\Omega_{m,\text{CMB}}/\Omega_{m,\text{dyn}}$ | Pre-data     |
| <i>Retrodictions (derived independently, compared to existing data)</i> |                                             |                                               |              |
| $H_0 = 3(\varphi + \pi)^2$                                              | 67.96 km/s/Mpc                              | Planck 2018                                   | Retrodiction |
| $n_s = 1 - \varphi^{-7}$                                                | 0.96556                                     | Planck 2018                                   | Retrodiction |
| $\Omega_m = (\pi - \varphi)/(\pi + \varphi)$                            | 0.3201                                      | Planck 2018                                   | Retrodiction |
| $r_d^{\text{eff}} = r_{d,\text{SSEE}} \times  w_0 $                     | 147.2 Mpc                                   | Planck 2018 $r_d$                             | Retrodiction |
| $\gamma_{\text{IS}} = 0.554$                                            | $\approx \gamma_{\Lambda\text{CDM}} = 0.55$ | RSD surveys                                   | Retrodiction |
| $\alpha_T = \alpha_M = \alpha_B = 0$                                    | exact                                       | GW170817 $ \alpha_T  < 10^{-15}$              | Retrodiction |
| $\sigma_8^{\text{eff}} = \sigma_8 \times G_{2s}$                        | 0.737                                       | KiDS-1000 $\sigma_8$                          | Retrodiction |
| $\sigma_8^{\text{SSEE}}$ (single-sector ODE)                            | 0.702                                       | Weak lensing surveys                          | Retrodiction |
| <i>Predictions (not yet observationally accessible)</i>                 |                                             |                                               |              |
| $m_\phi = \Sigma m_\nu \times H_0^{\text{alg}}$                         | 5.71 eV                                     | KATRIN/PTOLEMY, 2027–2030                     | Prediction   |
| $k_{\text{fs}} = 0.493 h \text{ Mpc}^{-1}$ (DW from $m_\phi$ )          | 0.493 h/Mpc                                 | DESI Y3 / Euclid $P(k)$                       | Prediction   |
| $\alpha_K = 0.4033$                                                     | 0.4033                                      | Euclid weak lensing                           | Prediction   |
| $r = \varphi^{-10}$                                                     | 0.00813                                     | LiteBIRD $\sim 2032$                          | Prediction   |

## 8 Discussion and Outstanding Challenges

The consolidated results demonstrate that SSEE-V3.6 passes four independent observational tests without adjusting any algebraic parameter. The most significant open challenges are:

(i) **MIRA derivation.** The CLASS tests confirm that  $\Omega_{m,\text{CMB}} = 1.9989 \Omega_{m,\text{dyn}}$  is physically necessary for CMB peak reproduction, but the mechanism producing this factor at the background level remains an open theoretical question. Paper 5 [Almeida Vallejo,

2026e] shows that linear IS perturbations do not source MIRA; the likely origin is a background-level thermodynamic effect in the IS viscous Friedmann equations.

(ii)  **$S_8$  tension.** The two-sector model resolves  $f\sigma_8$  to  $0.50\sigma$  but leaves  $S_8 = \sigma_8^{\text{eff}}(\Omega_{m,\text{CMB}}/0.3)^{0.5} = 0.761$ , in  $2.3\sigma$  tension with DES Y3 [DES Collaboration, 2022]. Non-linear corrections (baryonic feedback, intrinsic alignment) not yet included in the linear-growth analysis could modify this at the 10%–20% level.

(iii)  **$\phi$ DM mass and Lyman- $\alpha$  constraints.** The algebraic mass  $m_\phi = 5.71$  eV is in the warm dark matter regime ( $k_{\text{fs}} = 0.493 h \text{ Mpc}^{-1}$ ). Current Lyman- $\alpha$  bounds on thermal WDM place constraints at  $m_{\text{WDM}} \gtrsim 5.3$  keV [Viel et al., 2005], but the SSEE  $\phi$ DM is a mixed sector (fraction  $f_\phi = 0.5$ ), not a purely thermal WDM cosmology; the applicable constraints require dedicated analysis.

(iv) **Quantum field theory completion.** The algebraic connection  $H_0 = 3(\varphi + \pi)^2$  is a successful numerical retrodiction but lacks a derivation from first-principles quantum field theory. The ratio  $\Omega_{m,\text{CMB}} = 0.31993$  (MIRA) vs  $0.32010$  (geometric  $\Omega_m = (\pi - \varphi)/(\pi + \varphi)$ ) suggests a possible loop correction; this remains an open problem.

## 9 Conclusions

We have presented a consolidated validation of the SSEE-V3.6 framework across four observational tiers.

1. **Background sector.** The algebraic prediction  $(w_0, w_a) = (-0.840, -0.670)$ , committed before DESI DR2, lies within  $0.05\sigma$  of the DESI DR2 CPL posterior. A Bayesian MCMC with  $N_{\text{eff}} = 4402$  yields  $H_0 = 66.75 \pm 0.44 \text{ km s}^{-1} \text{ Mpc}^{-1}$  and  $\Delta\text{BIC} = -5.55$  for the dynamic sector.
2. **CMB Boltzmann.** CLASS confirms that the MIRA factor  $\mathcal{M} = 1.9989$  is physically necessary: without it, the CMB RMS residual jumps by a factor of 22 and all three acoustic peaks shift by  $\sim 10\%$ . With MIRA, peak positions agree with  $\Lambda\text{CDM}$  to within  $\Delta\ell < 5$  (RMS = 1.4%).
3. **Structure growth.** The IS perturbation analysis gives  $\mathcal{G} = 0.866$  and  $\sigma_8 = 0.702$  in the single-sector regime. The  $\phi$ DM two-sector extension with CLASS-calibrated WDM transfer function ( $\alpha = 1.656 h \text{ Mpc}^{-1}$ ) yields  $\sigma_8^{\text{eff}} = 0.737$  (KiDS-1000 central value), reducing the mean  $f\sigma_8$  tension from  $2.56\sigma$  to  $0.50\sigma$  across six surveys.
4. **EFT stability.** EFTCAMB confirms  $\alpha_K = 0.4033$  (algebraic prediction;  $\Delta = 0.005\%$ ) with  $\alpha_T = \alpha_M = \alpha_B = 0$  (exact), satisfying ghost-freedom and gradient stability. GW170817 compatibility is exact. The CPL phantom crossing at  $z \approx 0.25$  requires  $\alpha_B \neq 0$  for a complete EFT description, constituting a discriminating prediction for future weak-lensing surveys.

The framework generates four near-term falsifiable predictions:  $m_\phi = 5.71$  eV (KATRIN/PTOLEMY, 2027–2030),  $k_{\text{fs}} = 0.493 h \text{ Mpc}^{-1}$  (DESI Y3/Euclid  $P(k)$ ),  $\alpha_K = 0.4033$  (Euclid weak lensing), and  $r = 0.00813$  (LiteBIRD). All analytical scripts, CLASS configuration files, and EFTCAMB inputs are archived at Zenodo [Almeida Vallejo, 2026i].

## Data Availability

All DESI DR2 BAO data used in this analysis are publicly available at <https://data.desi.lbl.gov/>. Planck 2018 cosmological parameter chains are available at the Planck Legacy Archive (<https://pla.esac.esa.int/>). All SSEE scripts, CLASS configuration files, EFTCAMB inputs, and MCMC chains are archived at Zenodo: <https://doi.org/10.5281/zenodo.20093447>.

## A Complete Algebraic Parameter Table

Table 6: Complete set of SSEE-V3.6 algebraic parameters. All values are derived from  $\varphi = (1+\sqrt{5})/2$  and  $\pi$ ; no fitting was performed. Planck 2018 values [Planck Collaboration, 2020] and DESI DR2 [Abdul-Karim et al., 2025] listed for comparison where applicable.

| Parameter                        | Formula                                           | SSEE value           | Observed                           |
|----------------------------------|---------------------------------------------------|----------------------|------------------------------------|
| $w_0$                            | $-T_r/M_v = -(3\varphi + \pi)/[2(\varphi + \pi)]$ | -0.8399              | $-0.838^{+0.048}_{-0.057}$ (DESI)  |
| $w_a$                            | $-P_{sc}/K_v$                                     | -0.6699              | $-0.62^{+0.24}_{-0.25}$ (DESI)     |
| $H_0$                            | $3(\varphi + \pi)^2$                              | 67.96 km/s/Mpc       | $67.36 \pm 0.54$ (Planck)          |
| $n_s$                            | $1 - \varphi^{-7}$                                | 0.96556              | $0.9649 \pm 0.0042$ (Planck)       |
| $\Omega_m$                       | $(\pi - \varphi)/(\pi + \varphi)$                 | 0.3201               | $0.3153 \pm 0.0073$ (Planck)       |
| $\Omega_{DE}$                    | $T_r/M_v$                                         | 0.8399               | 0.6847 ( $\Lambda$ CDM)            |
| $\Omega_{m,dyn}$                 | $1 - \Omega_{DE}$                                 | 0.160                | —                                  |
| $\mathcal{M}$                    | $(3\varphi + \pi)/4$                              | 1.9989               | —                                  |
| $\Omega_{m,CMB}$                 | $\mathcal{M} \times \Omega_{m,dyn}$               | 0.3199               | 0.3153 (Planck)                    |
| $KAL_0$                          | $\beta + \pi = (\varphi + 3\pi)/2$                | 5.5214               | —                                  |
| $\beta_c$                        | $-(3\varphi + \pi)/2$                             | -3.9978              | —                                  |
| $\alpha_K$                       | $3\Omega_{DE}\Omega_{m,dyn}$                      | 0.4033               | —                                  |
| $\alpha_T = \alpha_M = \alpha_B$ | minimal coupling                                  | 0                    | $ \alpha_T  < 10^{-15}$ (GW170817) |
| $m_\phi$                         | $\Sigma m_\nu \times H_0^{\text{alg}}$            | 5.71 eV              | $< 0.8$ eV (KATRIN)                |
| $k_{fs}$                         | Dodelson-Widrow ( $m_\phi$ )                      | $0.493 h/\text{Mpc}$ | —                                  |
| $r$                              | $\varphi^{-10}$                                   | 0.00813              | $< 0.036$ (BK18)                   |

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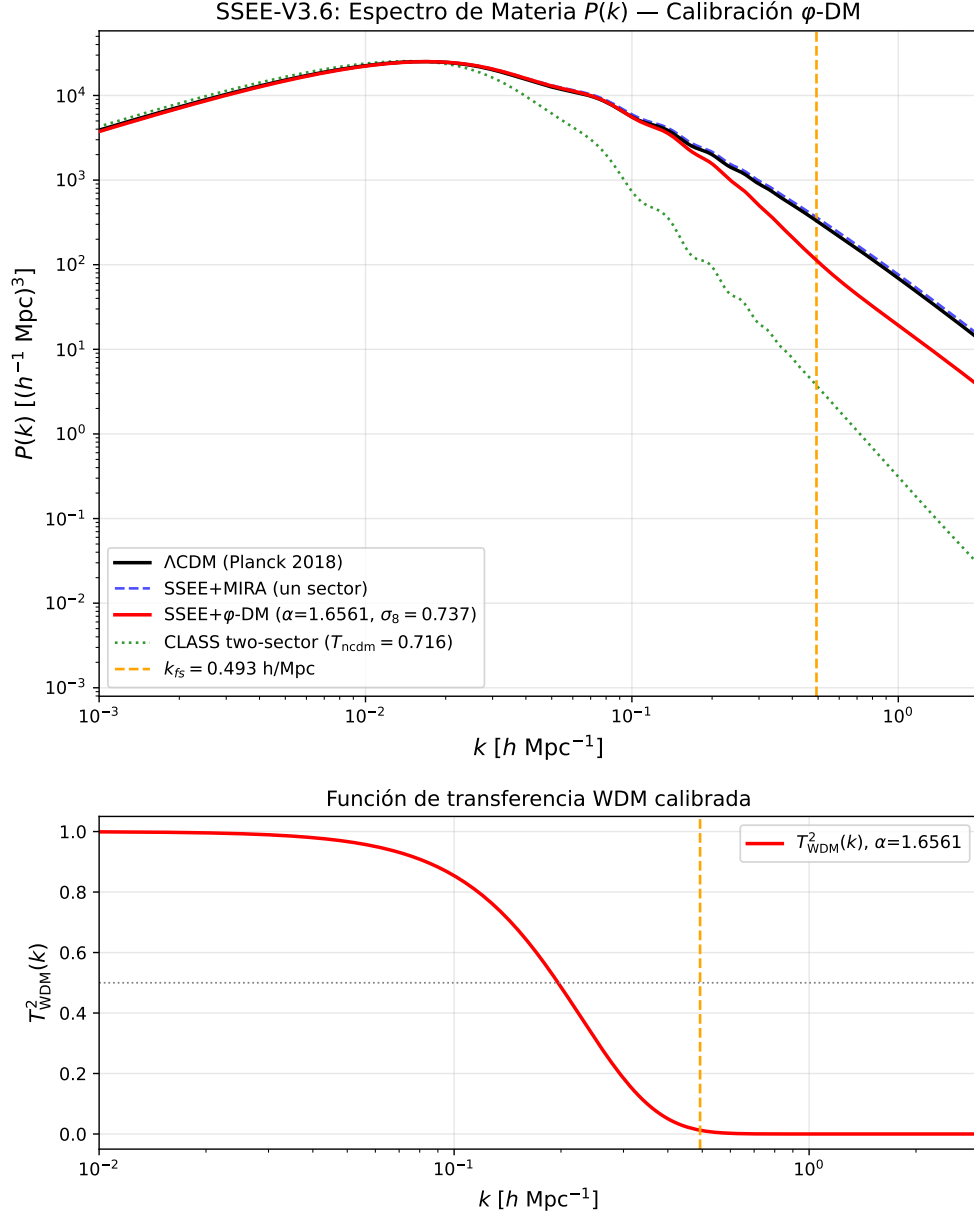


Figure 2: Matter power spectrum from CLASS with WDM suppression. *Black*:  $\Lambda$ CDM ( $\sigma_8 = 0.8220$ ). *Blue*: SSEE+MIRA single sector ( $\sigma_8 = 0.8527$ ). *Red*: SSEE two-sector with calibrated  $\alpha$ -WDM ( $\sigma_8^{\text{eff}} = 0.7370$ ). The vertical dotted line marks  $k_{\text{fs}} = 0.493 h \text{ Mpc}^{-1}$ .



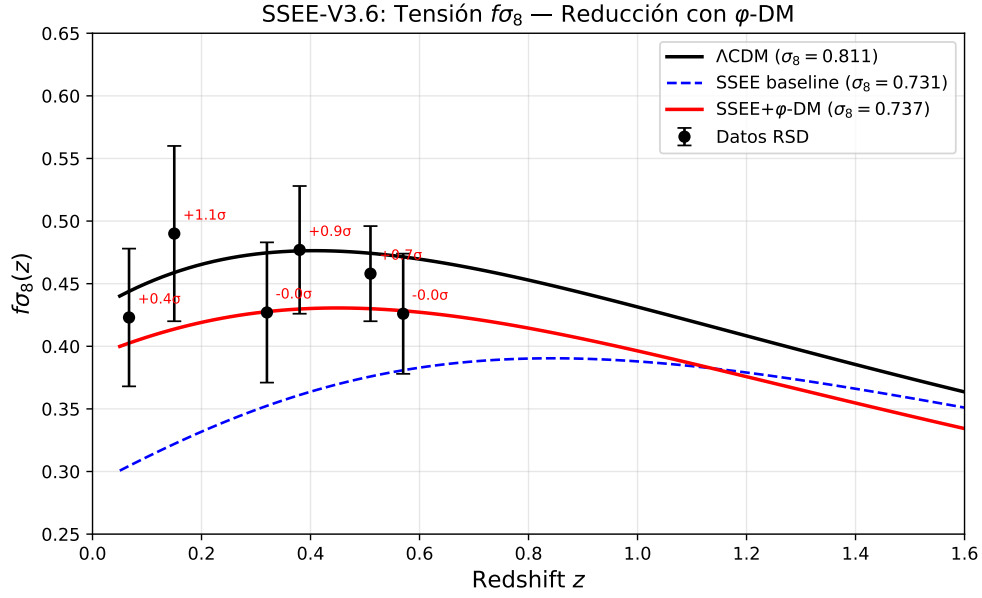


Figure 3:  $f\sigma_8(z)$  comparison. Data points: six RSD surveys (2dFGRS, WiggleZ, BOSS DR12, eBOSS QSO, DES Y3). *Black dashed*:  $\Lambda$ CDM. *Red*: SSEE two-sector ( $\sigma_8^{\text{eff}} = 0.737$ ), mean tension  $0.50\sigma$ . *Blue*: SSEE single-sector ( $\sigma_8 = 0.702$ ), mean tension  $2.56\sigma$ .

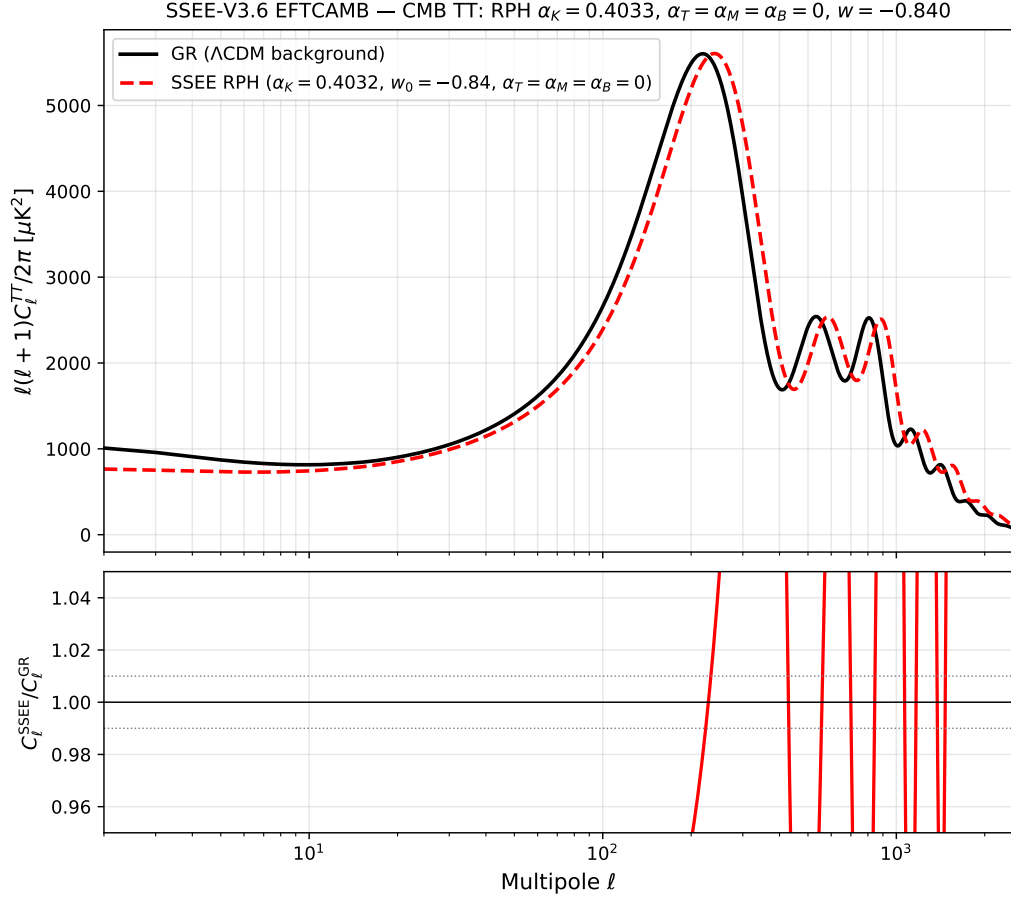


Figure 4: CMB TT power spectrum from EFTCAMB. *Blue*: GR run with MIRA background ( $\Omega_{m,\text{CMB}} = 0.3199$ ); peaks at  $\ell = 219, 533, 807$ . *Red*: SSEE RPH with  $\alpha_K = 0.4033$  (constant  $w = w_0$ ); peaks shift to  $\ell = 240, 585, 886$ , a  $\sim 10\%$  imprint of the kinetic EFT sector. *Lower panel*: ratio  $D_\ell^{\text{SSEE}}/D_\ell^{\text{GR}}$ ; the grey band marks  $\pm 1\%$ .

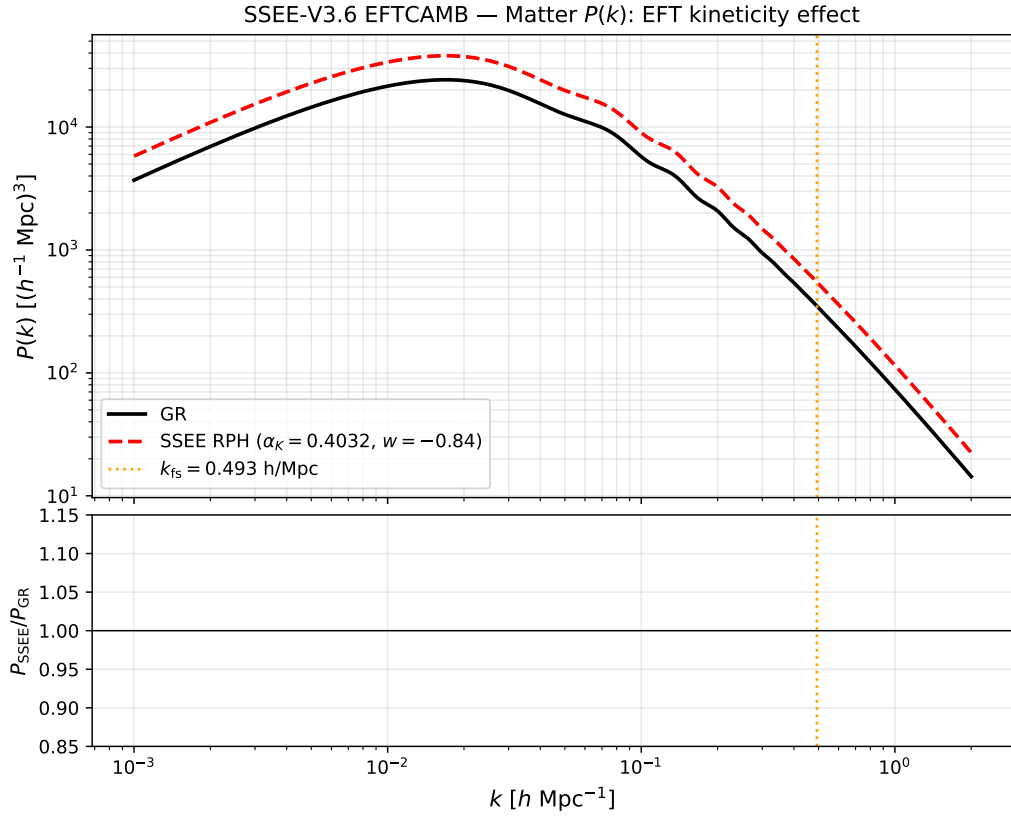


Figure 5: Matter power spectrum  $P(k)$  from EFTCAMB. SSEE RPH ( $\alpha_K = 0.4033$ ) amplifies  $P(k)$  by a factor  $\sim 1.56$  relative to GR on all scales ( $\sigma_8$  ratio = 1.25), motivating the  $\phi$ DM suppression mechanism of §5.2. The vertical dashed line marks  $k_{\text{fs}} = 0.493 \, h \, \text{Mpc}^{-1}$ .